INCH-POUND

MIL-A-83162A(USAF)
17 February 1989
SUPERSEDING
MIL-A-83162(USAF)
31 July 1968

MILITARY SPECIFICATION

AMPLIFIER, SYNCHRO, SIGNAL AM-6015/A

This specification is approved for use by, the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a synchro signal amplifier designated as AM-6015/A. The amplifier is used to provide increased torque to synchro torque receivers and minimize interaction between receiver loads, and as an isolation amplifier for control transformer loads.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

PPP-B-601	Wood, Boxes Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

Military

MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5990

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MIL-S-20708/5	Synchro Torque Receiver, Type 26V-11TR4C
MIL-S-20708/7	Synchro Control Transformer, Type 26V-11CT4D

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements Of
MIL-STD-471	Maintability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics And Utilization Of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Requirements For Reliability Program (System and Equipment
MIL-STD-2073-1	DOD Material Procedures For Development And Packaging Requirement
MS3102	Connector, Receptacle, Electric, Box Mounting

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (Attn: NPODS), 5801 Tabor Avenue, Philadelphia PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 General specification. The requirements of MIL-E-5400 shall apply as requirements of this specification with additions or exceptions to the paragraphs of MIL-E-5400, as identified by is similarity of paragraph headings, called out herein. When the two documents conflict, this specification shall govern.

3.2.1 Classification. The AM-6015/A amplifier shall be class 2 as defined in MIL-E-5400 with exceptions as specified herein.

3.2.2 Materials, parts, and processes

3.2.2.1 Approval of nonstandard parts and materials. Contract requirements for nonstandard parts and materials approval shall be category III as defined in MIL-E-5400.

3.2.2.2 Electrical connectors. The electrical receptacle for the amplifier shall be type MS3102R18-1P and shall be designated J-20.

3.2.2.3 Marking and identification

3.2.2.3.1 Label. The information specified in 3.3.2 shall appear on a label attached to the amplifier.

3.2.2.4 Electron tubes and semiconductor devices. Electron tubes shall not be used in the design of the amplifier.

3.2.3 Design and construction

3.2.3.1 Interference. The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of MIL-STD-461 and MIL-STD-462.

3.2.3.2 Reliability and maintainability

3.2.3.2.1 Reliability program. The contractor shall establish and conduct a reliability assurance program in accordance with MIL-STD-785.

3.2.3.2.2 Reliability. Reliability of the AM-6015/A amplifier shall be determined in accordance with MIL-STD-781. The amplifier shall have a minimum acceptable mean-time-between-failures (MTBF) of 5000 hours during its useful operating life.

3.2.3.2.3 Maintainability design requirements. Maintainability design requirements shall be demonstrated in accordance with MIL-STD-471.

3.2.3.3 Electrical overload protection. The amplifier shall withstand a short circuit to ground on either or both amplifier outputs for not less than 10 minutes. Upon removal of the short, the amplifier shall resume normal operation without

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resetting or replacing a fuse.

3.2.3.4 Service conditions (electrical). The amplifier shall be designated as a category B utilization equipment in accordance with MIL-STD-704.

3.2.3.4.1 Total input power. The required 115V, single phase, a-c power shall not exceed 300 va.

3.2.3.4.2 Warmup time. The amplifier shall not require warmup time.

3.2.3.4.3 Repeater excitation. The amplifier shall supply two modes of voltage excitation to the torque receivers. The primary mode shall be 26V, 400 Hz, at 25 watts maximum; this mode shall be supplied to pin B on the J-20 receptacle as specified in table I. The secondary mode shall be 86.5V, 400 Hz, at 10 watts maximum. When required, the secondary mode shall be obtainable, by internal wiring modifications, on pin B of the J-20 receptacle.

NOTE: All torque receivers driven from this amplifier should be excited from either the 115V, 400 Hz excitation to the AM-6015/A amplifier or by either of the two modes specified in 3.2.3.4.3.

3.2.3.5 Service conditions (environmental). Class 2 equipment requirements of MIL-E-5400 shall apply except as specified herein.

3.2.3.5.1 Temperature. The intermittent operation of the equipment shall be at 85°C.

3.3 Design and construction

3.3.1 Form factor. The form factor of the Amplifier, Synchro Signal AM-6015/A shall be in accordance with figure 1.

3.3.2 Receptacle pin designations. The J-20 amplifier receptacle shall have the pin designations as shown on table I.

3.4 Performance

3.4.1 Impedance

3.4.1.1 Input impedance. The input impedance shall be 10,000 ohms minimum.

3.4.1.2 Output impedance. The output impedance shall be 2 ohms maximum when working into an 100-ohm load with an output voltage of 10V rms at 400 Hz.

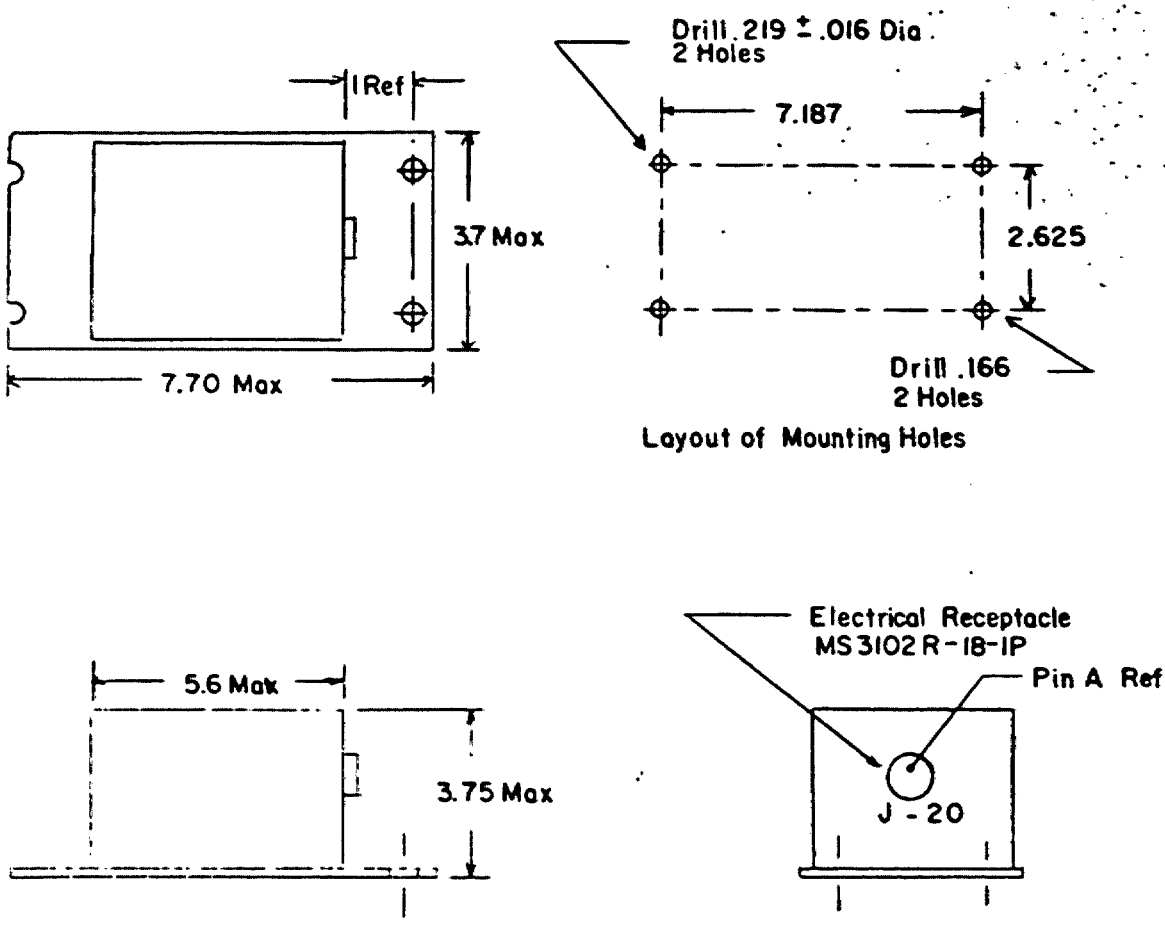
3.4.2 Load capability. The amplifier shall be capable of positioning four parallel connected torque receivers, type 26V-11-TR4C (MIL-S-20708/5B), or drive four control transformers, type 26V-11-CT4D (MIL-S-20708/5B).

3.4.3 Accuracy. The amplifier shall not introduce more than 0.15 degree of error into the torque receiver load specified in 3.4.2.

3.4.4 Multiple load isolation. The displacement of one of the multiple loads 20 degrees from the transmitter output shall not cause any other receiver load or transmitter to vary more than 0.25 degree.

3.5 Weight. The weight of the AM-6015/A amplifier shall not exceed 3 pounds.

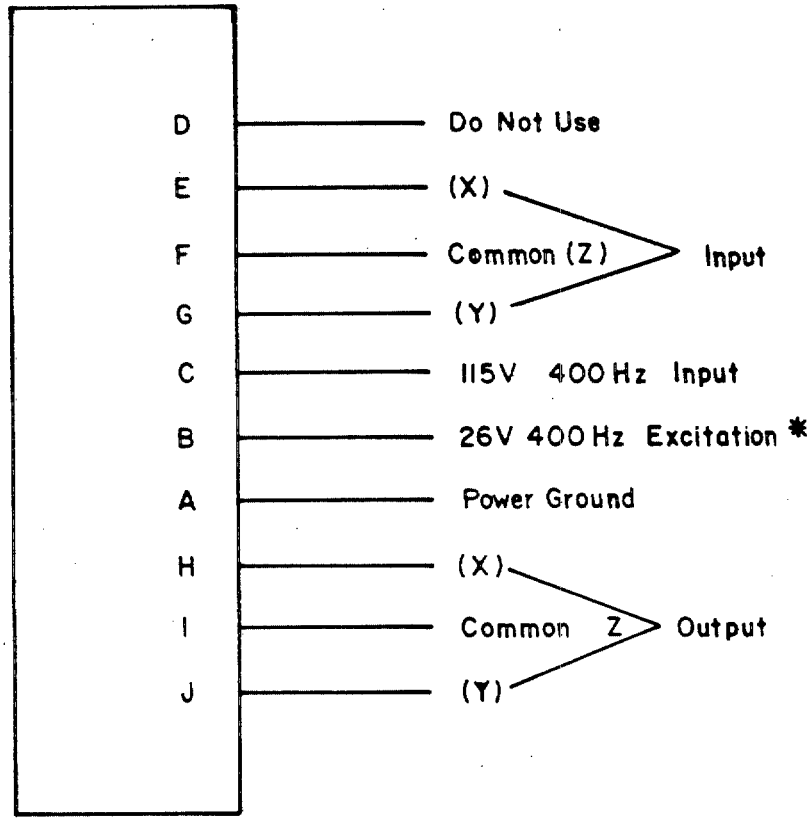
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Dimensions in Inches Unless Otherwise Noted
Tolerances: Decimals ±.005
Color: Black per FED-STD-595 (37038)

FIGURE 1. Outline of amplifier, synchro signal am-6015/a.

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* Note: 86.5V 400Hz may be obtained on pin B by internal wiring modification.

TABLE I. Pin designation of J-20 receptacle.

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3.6 Recycled and reclaimed materials. Recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the extend use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality assurance. The quality assurance provisions of MIL-E-5400 and the requirements specified herein shall be applicable to this specification. When the two documents conflict, this specification shall govern.

4.2.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, the amplifier shall be subjected to the tests under the following standard test conditions:

Temperature:	Room ambient (25°C ±5°C)
Altitude:	Normal ground
Vibration:	None
Humidity:	Room ambient up to 90 percent relative humidity
Input voltage:	115V ±1. 0V, 400 Hz, single phase

4.4 Preproduction inspections

4.4.1 Preproduction samples. Preproduction samples shall be representative of the production amplifier. They shall be tested under the conditions specified herein and at the point designated by the contract.

4.4.1.1 Testing instructions. The supplier shall conduct tests on the preproduction samples to determine that the design of the equipment meets the requirements

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specified herein. The supplier's tests shall be conducted in accordance with the test requirements specified herein.

4.4.1.1.1 Preproduction tests. One preproduction amplifier shall be subjected to the preproduction tests specified herein.

4.4.1.1.2 Preproduction reliability testing. Preproduction reliability testing shall be in accordance with test level F of MIL-STD-781. Quantities to be subjected to the preproduction reliability test and accept-reject criteria shall be in accordance with test plan XXVI of MIL-STD-781.

4.4.2 Preproduction tests. Preproduction tests shall consist of the tests specified under 4.6.

4.5 Quality conformance inspections. Quality conformance inspections shall consist of individual tests and sampling plans and tests.

4.5.1 Individual tests. Each amplifier shall be subjected to the following tests:

Tests	Applicable paragraph
	Examination of product
4.6.1	Burn-in
4.6.2	Power consumption
4.6.3	Voltage gain
4.6.4	Accuracy
4.6.5	Multiple load isolation
4.6.6	Overload
4.6.7	Repeater excitation voltage
4.6.8	Impedance
4.6.12	

4.5.2 Sampling plans and tests

4.5.2.1 Sampling plan A. The reliability production acceptance tests shall be conducted in accordance with the requirements of test level F of MIL-STD-781. The quantity to be tested shall be as specified in table II.

TABLE II. Sampling plan A and B quantity selection.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

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4.5.2.1.1 Accept-reject criteria. Accept-reject criteria shall be in accordance with test plan XXVII of MIL-STD-781.

4.5.2.2 Sampling plan B. Amplifiers selected at random in accordance with table II, and having passed the individual tests, shall be subjected to the following tests:

Tests	Applicable paragraph
Post-overload accuracy (The amplifier shall operate with a short for 10 minutes	4.6.7.1
Temperature and altitude	4.6.9.2
Vibration	4.6.9.3

4.5.2.2.1 Rejection and retest. When one amplifier selected from the production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The supplier shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.3 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of the amplifiers on hand or later produced shall not be made until it is determined that all amplifiers meet all the requirements of this specification.

4.5.4 Defects in amplifiers already accepted. The investigation of a test failure could indicate that defects may exist in amplifiers already accepted. If so, the supplier shall fully advise the procuring of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The amplifier shall be inspected to verify that the materials, design and construction, necessary mechancial measurements, marking, and workmanship comply with this specification.

4.6.2 Burn-in test. Each amplifier shall be operated under standard operating conditions at a maximum applied voltage of 118V ±1V, single phase and minimum applied frequency of 380 ±5 Hz for a period of 10 hours.

4.6.3 Power consumption test. The total power required for the amplifier shall not exceed 300 va starting and 100 va running. For the above measurement the amplifier shall be excited by an 11.8V transmitter (or equivalent) and loaded with four torque receivers, type 26V-11-TR4C (or equivalent).

4.6.4 Voltage gain test. The amplifier shall be delta connected to a 100-ohm load. With an input voltage of 10V rms, 400 Hz, the voltage gain of the two channels shall not differ by more than 0.2 percent.

Accuracy test. The amplifier shall be connected with a load of four torque receivers, type 26V-11-TR4C (or equivalent). With an input signal from a transmitter of known accuracy, the output data shall be measured every 30 degrees. The maximum error between transmitter data and the amplifier output shall be less

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than 0.15 degrees.

4.6.6 Multiple load isolation test. The amplifier shall be connected with a load off our torque receivers, type 26V-11-TR4C (or equivalent). One of the torque receivers shall be displaced 20 degrees. The heading output of the amplifier shall not deviate from the transmitter data by more than 0.25 degree.

4.6.7 Overload tests

4.6.7.1 Post-overload accuracy test. The amplifier shall be connected with a data transmitter (or equivalent) input. Both outputs shall be shorted to the ground and the amplifier allowed to operate for five minutes. Immediately following, the amplifier shall be connected as specified in 4.6.5 and rechecked for accuracy as specified in 4.6.5, except the measurement shall be every 90 degrees. The same accuracy tolerance shall apply.

4.6.7.2 Overload monitor. The amplifier shall be connected with a load of four torque receivers, type 26V-11-TR4C (or equivalent). One of the torque receivers shall be displaced 90 degrees. During the overload condition, the remaining three torque receivers shall osicllate between the heading input and the heading of the displaced torque receiver, indicating a torque receiver load malfunction.

4.6.8 Repeater excitation voltage test. The primary receiver excitation voltage of 26V ±1V, 400 Hz, at 26 watts maximum, shall be measured across a 26-ohm 50-watt resistor. The secondary receiver excitation voltage of 86.5V ±3.5V, 400 Hz, at 10 watts maximum, shall be measured across an 1000-ohm 10-watt resistor.

4.6.9 Environmental tests. Environmental tests shall be conducted in accordance with class 2 requirements of MIL-T-5422 with exceptions or additions as specified herein. When the two documents conflict, the requirements of this specification shall govern.

4.6.9.1 Completion of tests. After completion of each of the following test in MIL-T-5422, the amplifier shall meet the requirements of 4.6.5 and 4.6.6, unless otherwise specified:

a. Temperature and altitude: The tolerance specified in 4.6.5 shall be 0.25 degree; the tolerance specified in 4.6.6 shall be 0.30 degree."

b. Vibration, part I, curve I

c. Shock

d. Humidity

e. Salt fog

f. Explosion, procedure I

g. Dust (fine sand)

h. Fungus resistance

4.6.9.2 Temperature and altitude test

a. Step 1. The temperature of $-62^{o}C$ shall be maintained for 4 hours.

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b. Step 5. The temperature shall be 85°C and shall be maintained for 4 hours.

c. Steps 7 through 13 shall not be required.

4.6.9.3 Vibration test. Step 6 of part I shall not be required.

4.6.10 Interference test. The amplifier shall be subjected to interference tests in accordance with MIL-STD-461 and MIL-STD-462.

4.6.11 Maintainability demonstration. Maintainability demonstration shall be conducted in accordance with test method 5 of MIL-STD-471.

4.6.12 Impedance tests. The supplier shall conduct tests to determine compliance with the requirements specified in 3.4.1.1 and 3.4.1.2 herein.

4.7 Design data and material

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Amplifiers shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2 Drying. Amplifiers shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each amplifier shall be packaged in quantity unit packs of one each in accordance with Method IIc of MIL-P-116. Each amplifier shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the amplifier when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each amplifier shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of amplifiers shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Amplifiers packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Amplifiers packaged as specified in 5.1.1 shall be packed in

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shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged amplifier shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Amplifier, Synchro Signal AM-6015/A covered by this specification is intended to provide increased torque signals from the compass gyro standard heading transmitter to standard receiver (indicator, autopilot, navigation computer, and radar set).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Stock number

c. Preproduction samples and tests

(1) Number of preproduction samples

(2) Point of inspection

(3) Requirement for concurrent delivery of each sample. The contracting activity should be given at least 5 days prior notice when the preproduction tests are to be conducted so that they may be supervised or witnessed, if desired, by a Government representative.

d. Submission of test programs for approval (preproduction, acceptance, and reliability)

e. Selection of applicable levels or preservation and packaging, and packing.

f. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

6.3 Subject term (key work) listing.

Positioning
Receivers
Torque

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6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 71

Project no.
6615-F272

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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